PyeRDF Manual

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Installation and Usage Guide

1. Install Miniconda

As admin, open cmd. If on System32, exit and go to a user folder location.

```
curl https://repo.anaconda.com/miniconda/Miniconda3-latest-Windows-x86_64.exe -o .\miniconda.exe
start /wait "" .\miniconda.exe /S
del .\miniconda.exe
```

Search for *Anaconda Prompt* in Windows Explorer, open it, and create a virtual environment, here named venv:

```
conda create -n venv python==3.12.8
conda activate venv
```

If you prefer not to install Miniconda, download and install Python 3.12.8 directly from the official website:

```
https://www.python.org/downloads/windows/
```

2. Build and Install the Package

Download the ZIP from GitHub and extract it. Then, navigate into the extracted folder. For example, if extracted under your user directory:

```
cd "C:\Users\seccolev\PyeRDF-main"
```

Inside the folder, run:

```
python -m pip install --upgrade pip build
python -m build
```

This will create a dist folder, and inside it there will be a `.whl` file. Use pip to install this file.

```
pip install dist/phd-0.1.0-py3-none-any.whl
```

The installation should be now complete with all the dependencies.

3. Running the Scripts

(1) Azimuthal Integration

```
python azim_integ.py
```

This script finds the center and azimuthally averages the 2D diffraction pattern. Choose a `.tif` file. If the center is not found (common with beamstopper), the script will ask for a new threshold (0–255). Lower values often work better. When the center is found, this graph will display. Here you can observe if the found center is right (1). Close this window. After closing, the found center and threshold will be displayed in cmd and you will be prompted (2).

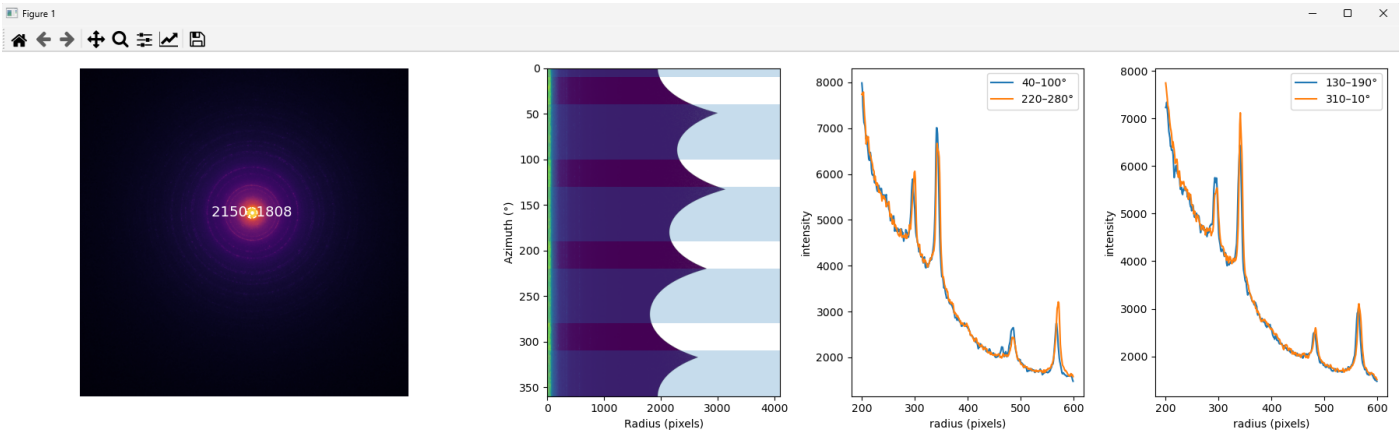

Figure 1: Center Evalutation

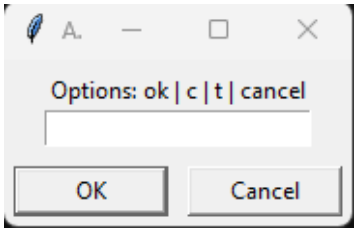

Figure 2: Center fine-tuning prompt.

Type c and you will be able to enter the center manually, e.g.: 2150, 1908. Type "t" and and choose a new threshold. Type ok to accept the center. After the center is accepted, it will export a single-column `.csv` file of intensities.

(2) Calibration

```
python calibrate.py
```

Use a polycrystalline gold diffraction pattern. After finding the center, a 1D profile will appear with the detected peaks.

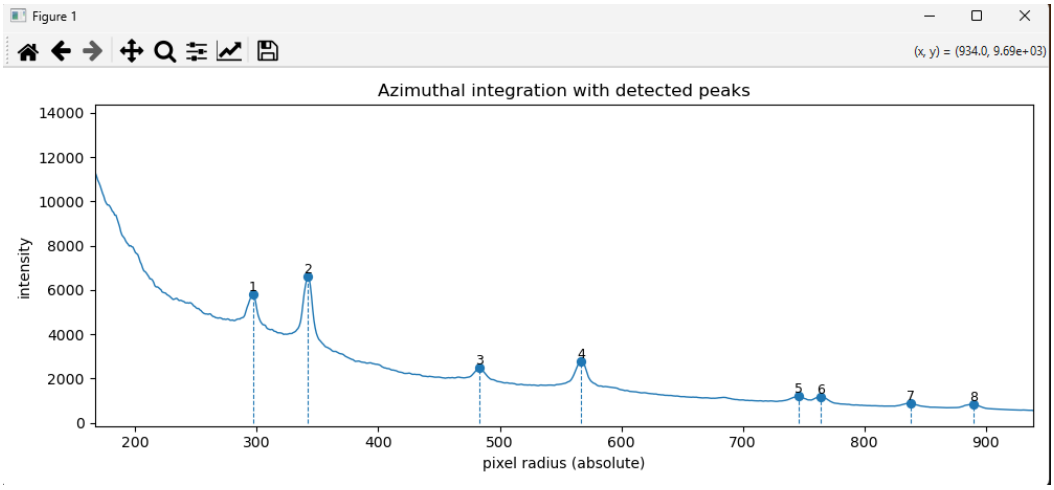

Figure 3: Gold calibration.

Select the first four main peaks by typing their indices separated by commas, e.g.:

```
1,2,3,4
```

The script outputs the calibration in **q**-scale (`dq`) and prints it on cmd. It also lets you select a `.csv` or `.txt` file to store it in the first row.

(3) PDF Calculation

```
python erdf.py
```

Choose a `.csv` file containing a single-column intensity profile, where the first row is the calibration factor (`dq`). A parameter window will appear for fitting and PDF calculation settings.

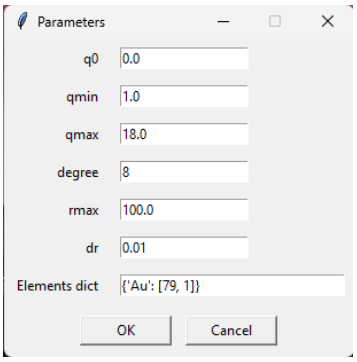

Figure 4: Fitting Parameters

Q_0 will assume that the first data point in the `.csv` file corresponds to the provided value. Q_{min} and Q_{max} will select the data interval to fit based on the calibration provided. **degree** corresponds to the polynomial degree used in the reduced structure function $F(Q)$ generation. The fitting methodology was based on the description in the *DISCUS* manual (`https://github.com/tproffen/DiffuseCode`), which follows the same empirical approach used in *PDFGetX3*. `rmax` and `dr` are used in the PDF calculation.

Unfortunately, a drop-down menu for selecting elements has not yet been implemented. At the moment, you need to provide them in Python dictionary format. The first number is the atomic number, and the second is the number of atoms in the molecule or unit cell. Examples:

```
{'Fe': [26, 3], 'O': [8, 4]}
{'Pb': [82, 1], 'S': [16, 1]}
```

The script will then show the calibrated intensity with the atomic scattering factors profile fit, the $F(Q)$ and the $G(r)$. After closing the graph it will prompt to select a folder, in which the data displayed in the graphs will be saved as `.csv` two-column files.